Lecture Scribe Transformation

L12_S1_A.pdf
PDF

You are a course expert in computation probability and act as a scholar.
I am providing you lecture PowerPoint presentation. All you have to do is to convert it into a proper and a detailed version of it in lecture scribe format. The lecture scribe should contain:
Make sure to identify and cover the main concepts, definitions, formulae, diagrams and examples
Infer explanations from each topic and give a detailed note on it.
For each slide/topic, generate:
Topic heading and highlight the important topics
Explain the concepts in the simple language and easy words with proper examples
Include derivation steps
Explain logic in simple words
Include all the applications given in the ppt
Split down the complex concepts into step by step and simpler explanations
Do explain using conceptual approach and mathematic approach
If you find any inconsistency, solve it and generate the most accurate explanation
Give summary and make sure to include important topics
Give the formulas sheet
Do not assume prior knowledge
Do not skip any slide and topic

Perfect, I've carefully studied your uploaded lecture slides **"CSE400 – Lecture L11_12: Transformation of Random Variables"** and converted **every slide and handwritten derivation** into a clean, structured **lecture scribe** with:

- Simple language (no prior knowledge assumed)
- Conceptual + mathematical explanations
- Step-by-step derivations

- All examples (sin transform, sum of RVs, Gaussian case, exponential case, difference, ratio, radius example)
- Diagrams explained in words (regions, limits, geometry)
- Final summary + formula sheet

All content below is derived directly from your slides.

CSE400 – Fundamentals of Probability in Computing

Lecture L11–L12: Transformation of Random Variables
Instructor: Dhaval Patel, PhD
Topic: Transformation of Random Variables & Functions of Two Random Variables

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1. Why Do We Need Transformation of Random Variables?

In probability, many times:

- We **know** a random variable X and its PDF/CDF
- But we are **interested in another variable** Y that depends on X
  For example:
  - $Y = g(X) = \sin(\pi X/2)$
  - $Z = X + Y$
  - $Z = X - Y$
  - $Z = X / Y$
  - $R = \sqrt{(X^2 + Y^2)}$

So the main question is:

If X (or X, Y) are known, how do we find the **distribution (CDF/PDF)** of the new variable?

This process is called **Transformation of Random Variables**.

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2. Case 1: Transformation of One Random Variable

Let:

- X be a continuous random variable with known PDF $f_x(x)$ and CDF $F_x(x)$

- Define a new variable:
  $Y = g(X)$

Goal: Find $\mathbf{F_Y(y)}$ and $\mathbf{f_Y(y)}$

Step 1: Use the CDF Method

By definition:
$F_Y(y) = P(Y \leqslant y)$

Since $Y = g(X)$:
$F_Y(y) = P(g(X) \leqslant y)$

If g(.) is **monotonic increasing**, then:
$g(X) \leqslant y \Leftrightarrow X \leqslant g^{-1}(y)$

So:
$F_Y(y) = P(X \leqslant g^{-1}(y)) = F_x(g^{-1}(y))$

Step 2: Differentiate to Get PDF

$f_Y(y) = d/dy\ [\ F_Y(y)\ ]$
$= d/dy\ [\ F_x(g^{-1}(y))\ ]$

Using chain rule:
$f_Y(y) = f_x(g^{-1}(y)) \cdot |\ d/dy\ (g^{-1}(y))\ |$

So the **main formula** is:

$f_Y(y) = f_x(x) \cdot |\ dx/dy\ |$ evaluated at $x = g^{-1}(y)$

If g(.) is Decreasing?

Then:
$F_Y(y) = P(Y \leqslant y) = P(X \geqslant g^{-1}(y)) = 1 - F_x(g^{-1}(y))$

Differentiate again to get PDF, and the same **absolute Jacobian rule** appears.

---

3. Example 1: $Y = \sin(\pi X / 2)$, $X \sim \text{Uniform}(-1, 1)$

Given:

$f_x(x) = 1/2$, for $-1 < x < 1$

0 otherwise

Define:

$Y = \sin(\pi X / 2)$

Step 1: Find the Inverse

$y = \sin(\pi x / 2)$

So:

$x = (2/\pi) \sin^{-1}(y)$

Step 2: Compute dx/dy

$dx/dy = (2/\pi) \cdot 1/\sqrt{(1 - y^2)}$

Step 3: Apply Transformation Formula

$f_Y(y) = f_x(x) \cdot |dx/dy|$

$= (1/2) \cdot (2/\pi) \cdot 1/\sqrt{(1 - y^2)}$

$= 1 / (\pi \sqrt{(1 - y^2)})$

Valid for: $-1 < y < 1$

So:

$f_Y(y) = 1 / (\pi \sqrt{(1 - y^2)})$, $-1 < y < 1$

0 otherwise

This is a **U-shaped distribution** (as sketched in the slide).

---

4. Case 2: Function of Two Random Variables

Now let:
$Z = g(X, Y)$

We want:

- $F_z(z) = P(Z \leqslant z)$
- $f_z(z) = d/dz [F_z(z)]$

General idea:

$F_z(z) = P(g(X, Y) \leqslant z)$
$= \iint$ over region $\{g(x,y) \leqslant z\}$ $f_{X,Y}(x,y)\, dx\, dy$

Then:
$f_z(z) = d/dz \iint_{\{g(x,y) \leqslant z\}} f_{X,Y}(x,y)\, dx\, dy$

To differentiate an integral with variable limits, we use **Leibniz Rule**.

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5. Leibniz Rule (From Slides)

If:
$G(x) = \int$ from $a(x)$ to $b(x)$ of $h(x, y)\, dy$

Then:
$dG/dx = h(x, b(x)) \cdot db/dx - h(x, a(x)) \cdot da/dx + \int$ from $a(x)$ to $b(x)$ of $\partial h/\partial x\, dy$

This rule is used to differentiate integrals where **limits depend on the variable**.

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6. Main Example: Z = X + Y

Goal:
Find PDF of Z

Step 1: Write CDF

$F_z(z) = P(Z \leqslant z) = P(X + Y \leqslant z)$

This means we integrate joint PDF over the region:
$x + y \leqslant z$

Graphically (as in slides), this is a half-plane under the line:

$y = z - x$

So:

$F_z(z) = \iint_{x + y \leqslant z} f_{X,Y}(x, y)\, dx\, dy$

Step 2: Differentiate w.r.t z

Using Leibniz rule, we get:

$f_z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx$

If X and Y are Independent

Then:

$f_{X,Y}(x, y) = f_x(x)\, f_y(y)$

So:

$f_z(z) = \int_{-\infty}^{\infty} f_x(x)\, f_y(z - x)\, dx$

This is called the **Convolution Integral**.

---

7. Special Case: X ~ N(0,1), Y ~ N(0,1), Independent

Given:

$f_x(x) = (1/\sqrt{(2\pi)})\, e^{-x^2/2}$

$f_y(y) = (1/\sqrt{(2\pi)})\, e^{-y^2/2}$

Then:

$f_z(z) = \int (1/\sqrt{(2\pi)})\, e^{-x^2/2} \cdot (1/\sqrt{(2\pi)})\, e^{-(z-x)^2/2}\, dx$

Combine exponents and simplify (as shown in slides):

You finally get:

$f_z(z) = 1 / \sqrt{(4\pi)} \cdot e^{-z^2/4}$

This is a **Normal distribution** with:

Mean = 0

Variance = 2

So:

$Z = X + Y \sim N(0, 2)$

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8. Another Case: X, Y are Exponential(λ)

Given:

$f_x(x) = \lambda e^{-\lambda x}, x \geqslant 0$

$f_Y(y) = \lambda e^{-\lambda y}, y \geqslant 0$

Then:

$f_z(z) = \int f_x(x) f_Y(z - x) dx$

But constraints:

$x \geqslant 0$ and $z - x \geqslant 0 \Rightarrow 0 \leqslant x \leqslant z$

So:

$f_z(z) = \int_0^z \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(z-x)} dx$

$= \lambda^2 e^{-\lambda z} \int_0^z dx$

$= \lambda^2 z e^{-\lambda z}, z \geqslant 0$

This is a **Gamma distribution with shape 2**.

---

9. Example: Z = X − Y

CDF:

$F_z(z) = P(X - Y \leqslant z)$

Region:

$x - y \leqslant z \Leftrightarrow y \geqslant x - z$

So integrate joint PDF over that region and then differentiate:

$f_z(z) = d/dz [F_z(z)]$

The **same procedure** as $Z = X + Y$, only the region changes.

---

10. Example: $Z = X / Y$

CDF:
$F_Z(z) = P(X / Y \leqslant z)$

Now we must consider **cases**:

- If $Y > 0$, inequality stays same
- If $Y < 0$, inequality reverses

So the plane is split into two regions (as shown in slide with colored areas), and integration is done **piecewise**.

---

11. Example: $R = \sqrt{(X^2 + Y^2)}$

This is a **radius** in 2D plane.

Meaning:

- (X, Y) is a random point
- R is the distance from origin

So:
$F_R(r) = P(\sqrt{(X^2 + Y^2)} \leqslant r) = P(X^2 + Y^2 \leqslant r^2)$

This is the probability that (X, Y) lies **inside a circle of radius r**.

So we integrate joint PDF over a **circular region**.

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12. Big Picture Summary

You learned:

- How to transform **one random variable**: $Y = g(X)$
- Use **CDF method** + derivative
- Main formula:
  $f_Y(y) = f_X(g^{-1}(y)) \cdot |dx/dy|$
- How to transform **two random variables**: $Z = g(X, Y)$

- Use:
  $F_z(z) = \iint_{g(x,y) \leqslant z} f_{X,Y}(x,y)\, dx\, dy$
- Then differentiate using **Leibniz rule**
- For sum:
  $f_z(z) = \int f_x(x)\, f_y(z - x)\, dx$ (Convolution)
- Applications:
  - Sum of Gaussians → Gaussian
  - Sum of exponentials → Gamma
  - Difference, ratio, radius, etc.

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13. Formula Sheet (Quick Revision)

1. One variable transform $Y = g(X)$:

$f_Y(y) = f_x(g^{-1}(y)) \cdot |\, dx/dy \,|$

2. Two variables $Z = X + Y$:

$f_z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx$

If independent:

$f_z(z) = \int_{-\infty}^{\infty} f_x(x)\, f_y(z - x)\, dx$

3. CDF method:

$F_z(z) = \iint_{g(x,y) \leqslant z} f_{X,Y}(x,y)\, dx\, dy$
$f_z(z) = d/dz\, [F_z(z)]$

4. Leibniz Rule:

$d/dx \int_{a(x)}^{b(x)} h(x,y)\, dy$
$= h(x,b(x))\, b'(x) - h(x,a(x))\, a'(x) + \int_{a(x)}^{b(x)} \partial h/\partial x\, dy$

If you want, I can now:

- Rewrite this into **exam-ready notes**, or
- Solve **practice problems** step by step like we did for CSE400 earlier, or
- Convert this into **LaTeX lecture notes** for you.

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CSE400 – Fundamentals of Probability in Computing

Lecture L11–L12: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Topic: Transformation of Random Variables & Functions of Two Random Variables

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Slide 1: What is This Lecture About?

Main Idea

In many problems:

- We **already know** a random variable X.
- We know its **PDF** $f_x(x)$ and **CDF** $F_x(x)$.
- But we are **interested in a new variable** formed from X or from (X, Y).

Examples shown in the slide:

- Y = g(X) (a function of one random variable)
- Z = g(X, Y) such as:
  - Z = X + Y
  - Z = X − Y
  - Z = X / Y
  - Z = X · Y

Goal of the Lecture

If X (or X, Y) is known:

- How do we find **$F_Y(y)$, $f_Y(y)$** or **F_z(z), f_z(z)**?
  This process is called **Transformation of Random Variables**.

Slide 2: Outline of the Lecture

1. Transformation of Random Variables (single variable case)
2. Function of Two Random Variables (joint transformation)
3. Illustrative Example: Z = X + Y (full derivation)

The handwritten notes also highlight:

- g(.) can be **monotonic increasing** or **monotonic decreasing**
- The limits and sign matter when transforming.

Slide 3: Case 1 — Transformation of One Random Variable

Problem Setup

Let:

- X be a continuous random variable.
- Its PDF $f_x(x)$ and CDF $F_x(x)$ are known.
- Define a new variable:
  $Y = g(X)$

We want:

- The **CDF** $F_Y(y)$
- The **PDF** $f_Y(y)$

Step 1: Use the CDF Method (Very Important Idea)

By definition:
$F_Y(y) = P(Y \leqslant y)$

Since $Y = g(X)$:
$F_Y(y) = P(g(X) \leqslant y)$

Now, if g(.) is **monotonic increasing**:

$g(X) \leqslant y$ means $X \leqslant g^{-1}(y)$

So:
$F_Y(y) = P(X \leqslant g^{-1}(y)) = F_x(g^{-1}(y))$

This step is exactly what is drawn in the slide using graphs of $y = g(x)$ and $x = g^{-1}(y)$.

Step 2: Differentiate to Get the PDF

We know:
$f_Y(y) = d/dy\ [\ F_Y(y)\ ]$

So:
$f_Y(y) = d/dy\ [\ F_x(g^{-1}(y))\ ]$

Using chain rule:
$f_Y(y) = f_X(g^{-1}(y)) \cdot d/dy\ [\ g^{-1}(y)\ ]$

Taking absolute value (to handle increasing/decreasing cases):

Final Formula:
$f_Y(y) = f_X(x) \cdot |\ dx/dy\ |$ evaluated at $x = g^{-1}(y)$

Slide 4: If g(x) is Decreasing

The slide shows:

$F_Y(y) = P(Y \leqslant y) = P(g(X) \leqslant y)$

If g is **decreasing**, then:
$g(X) \leqslant y$ means $X \geqslant g^{-1}(y)$

So:
$F_Y(y) = P(X \geqslant g^{-1}(y)) = 1 - F_X(g^{-1}(y))$

Differentiate:
$f_Y(y) = -\ f_X(g^{-1}(y)) \cdot d/dy\ [\ g^{-1}(y)\ ]$

Taking absolute value again gives the **same general formula**:

$f_Y(y) = f_X(g^{-1}(y)) \cdot |\ dx/dy\ |$

So the **same formula works for both increasing and decreasing** functions.

Slide 5: Example 1 — Y = sin(πX / 2), X ~ Uniform(−1, 1)

Given

X is uniformly distributed in (−1, 1):

$f_X(x) = 1/2$, for $-1 < x < 1$
0 otherwise

Define:
$Y = \sin(\pi X / 2)$

We must find $\mathbf{f_Y(y)}$.

Step 1: Find the Inverse Function

$y = \sin(\pi x / 2)$

So:
$x = (2/\pi) \sin^{-1}(y)$

Step 2: Compute dx/dy

$dx/dy = (2/\pi) \cdot 1 / \sqrt{(1 - y^2)}$

Step 3: Apply Transformation Formula

$f_Y(y) = f_X(x) \cdot |dx/dy|$

$= (1/2) \cdot (2/\pi) \cdot 1 / \sqrt{(1 - y^2)}$

$= 1 / (\pi \sqrt{(1 - y^2)})$

Domain:
Since sine output is between −1 and 1:

Final Answer:
$f_Y(y) = 1 / (\pi \sqrt{(1 - y^2)})$, for $-1 < y < 1$
0 otherwise

The slide also sketches the **U-shaped curve** of this PDF.

Slide 6: Case 2 — Function of Two Random Variables

Now define:
$Z = X + Y$

The slide asks to find:

(i) PDF of Z, $f_z(z)$

(ii) $f_z(z)$ if X and Y are independent

(iii) If $X \sim N(0,1)$, $Y \sim N(0,1)$, show $Z \sim N(0,2)$

(iv) If X, Y are exponential with parameter λ, find $f_z(z)$

Slide 7: CDF of Z = X + Y (Geometric View)

Start from definition:

$F_z(z) = P(Z \leqslant z) = P(X + Y \leqslant z)$

This means:

All (x, y) points such that $x + y \leqslant z$

On the x–y plane, this is the **half-plane below the line**:

$y = z - x$

So:

$F_z(z) = \iint_{\{x + y \leqslant z\}} f_{X,Y}(x, y)\, dx\, dy$

The slide shows this shaded region clearly.

Slide 8: Leibniz Rule (Needed for Differentiation)

If:

$G(x) = \int$ from a(x) to b(x) of h(x, y) dy

Then:

$dG/dx = h(x, b(x)) \cdot db/dx - h(x, a(x)) \cdot da/dx + \int$ from a(x) to b(x) $\partial h/\partial x\, dy$

We use this because **the limits depend on z** in $F_z(z)$.

Slide 9: Deriving the PDF of Z

Start from:

$F_z(z) = \iint_{\{x + y \leqslant z\}} f_{X,Y}(x, y)\, dx\, dy$

Differentiate w.r.t z using Leibniz rule:

You get:

$f_z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx$

This is the **general result** for $Z = X + Y$.

Slide 10: If X and Y Are Independent (Convolution)

If X and Y are independent:

$f_{X,Y}(x, y) = f_x(x)\, f_y(y)$

So:

$f_z(z) = \int_{-\infty}^{\infty} f_x(x)\, f_y(z - x)\, dx$

This integral is called the **Convolution Integral**.

Slide 10–11: Special Case — Sum of Two Standard Gaussians

Given:

$X \sim N(0,1)$, $Y \sim N(0,1)$, independent

So:

$f_x(x) = (1/\sqrt{(2\pi)})\, e^{-x^2/2}$

$f_y(y) = (1/\sqrt{(2\pi)})\, e^{-y^2/2}$

Then:

$f_z(z) = \int (1/\sqrt{(2\pi)})\, e^{-x^2/2} \cdot (1/\sqrt{(2\pi)})\, e^{-(z-x)^2/2}\, dx$

Combine exponents and complete the square (as shown step-by-step in the slide):

Final result:

$f_z(z) = 1 / \sqrt{(4\pi)} \cdot e^{-z^2/4}$

This is a **Normal distribution** with:

Mean = 0

Variance = 2

So:

$Z = X + Y \sim N(0, 2)$

Slide 12: X, Y Are Exponential(λ)

Given:

$f_x(x) = \lambda e^{-\lambda x}, x \geqslant 0$

$f_y(y) = \lambda e^{-\lambda y}, y \geqslant 0$

Using convolution:

$f_z(z) = \int f_x(x) f_y(z - x) dx$

But constraints:

$x \geqslant 0$ and $z - x \geqslant 0 \Rightarrow 0 \leqslant x \leqslant z$

So:

$f_z(z) = \int_0^z \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(z-x)} dx$

$= \lambda^2 e^{-\lambda z} \int_0^z dx$

$= \lambda^2 z e^{-\lambda z}, z \geqslant 0$

This is a **Gamma distribution (shape = 2)**.

Slide 13: Example — Z = X − Y

CDF:

$F_z(z) = P(Z \leqslant z) = P(X - Y \leqslant z)$

Rewrite:

$X - Y \leqslant z \Leftrightarrow Y \geqslant X - z$

So integrate joint PDF over that region in the x–y plane (as shaded in the slide).

Then:

$f_z(z) = d/dz [F_z(z)]$

The **procedure is exactly the same** as for Z = X + Y, only the region changes.

Slide 14: Example — Z = X / Y and R = $\sqrt{(X^2 + Y^2)}$

Case 1: Z = X / Y

CDF:

$F_z(z) = P(X / Y \leqslant z)$

Now we must split into cases:

- If $Y > 0$: inequality stays the same
- If $Y < 0$: inequality **reverses**

So the plane is divided into two regions (as drawn in the slide), and integration is done **piecewise**.

Case 2: $R = \sqrt{(X^2 + Y^2)}$

Interpretation:

- (X, Y) is a random point in 2D
- R is the **distance from origin**

So:
$F_R(r) = P(\sqrt{(X^2 + Y^2)} \leqslant r) = P(X^2 + Y^2 \leqslant r^2)$

This means:
Probability that (X, Y) lies **inside a circle of radius r**

So:
$F_R(r) = \iint$ over circle of radius r $f_{X,Y}(x, y)\, dx\, dy$

Then:
$f_R(r) = d/dr\, [\, F_R(r)\,]$

Final Summary (From the Whole Lecture)

You learned:

1. How to transform **one random variable**: $Y = g(X)$
   - Use CDF method
   - Differentiate to get PDF
   - Formula:
     $f_Y(y) = f_X(g^{-1}(y)) \cdot |dx/dy|$
2. How to transform **two random variables**: $Z = g(X, Y)$
   - First find:
     $F_Z(z) = \iint_{\{g(x,y) \leqslant z\}} f_{X,Y}(x,y)\, dx\, dy$

- Then:

  $f_Z(z) = d/dz\ [F_Z(z)]$ using Leibniz rule

3. For sum $Z = X + Y$:
   - General:

     $f_Z(z) = \int f_{X,Y}(x, z - x)\, dx$
   - If independent:

     $f_Z(z) = \int f_X(x)\, f_Y(z - x)\, dx$ (Convolution)
4. Applications covered:
   - $Y = \sin(\pi X/2)$ with uniform X
   - $Z = X + Y$ (general, Gaussian case, exponential case)
   - $Z = X - Y$
   - $Z = X / Y$
   - $R = \sqrt{(X^2 + Y^2)}$

Formula Sheet (Quick Revision)

1. One variable transform:

   $f_Y(y) = f_X(g^{-1}(y)) \cdot |\, dx/dy\, |$
2. Two variable transform:

   $F_Z(z) = \iint_{g(x,y) \leqslant z} f_{X,Y}(x,y)\, dx\, dy$

   $f_Z(z) = d/dz\ [F_Z(z)]$
3. Sum of RVs:

   $f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx$

If independent:

$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)\, f_Y(z - x)\, dx$ (Convolution)

4. Leibniz Rule:

   $d/dx \int_{a(x)}^{b(x)} h(x,y)\, dy$

   $= h(x,b(x))\, b'(x) - h(x,a(x))\, a'(x) + \int_{a(x)}^{b(x)} \partial h/\partial x\, dy$

If you want, I can now:

- Convert this into **clean exam-ready notes**, or
- Solve **extra practice problems** from this topic step by step, or
- Generate this in **LaTeX lecture-scribe format** for submission/printing.

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skipped, plus a final summary and formula sheet.
All content below is derived directly from your file.

CSE400 – Fundamentals of Probability in Computing

Lecture L11–L12: Transformation of Random Variables
Instructor: Dhaval Patel, PhD

Slide 1: Introduction to Transformation of Random Variables

Main Idea

In probability, many times:

- We **already know** a random variable X.
- We know its **PDF** $f_x(x)$ and **CDF** $F_x(x)$.
- But we are interested in a **new variable** made from X (or from X and Y).

Examples shown in the slide:

- $Y = g(X) \rightarrow$ function of one random variable
- $Z = g(X, Y)$ such as:
  - $Z = X + Y$
  - $Z = X - Y$
  - $Z = X / Y$
  - $Z = X \cdot Y$

Goal of This Lecture

If X (or X, Y) is known:

- How do we find **$F_Y(y)$, $f_Y(y)$** or **F_z(z), f_z(z)**?

This process is called **Transformation of Random Variables**.

Slide 2: Outline of the Lecture

1. Transformation of Random Variables (single variable case)
2. Function of Two Random Variables (joint transformations)
3. Illustrative Example: Z = X + Y (full detailed derivation)

The handwritten notes also highlight:

- The function g(.) can be **monotonically increasing** or **monotonically decreasing**
- The **limits and sign** matter while transforming

Slide 3: Case 1 — Transformation of One Random Variable

Problem Setup

Let:

- X be a continuous random variable.
- Its PDF $f_x(x)$ and CDF $F_x(x)$ are known.
- Define a new variable:
  $Y = g(X)$

We want:

- The **CDF** $F_Y(y)$
- The **PDF** $f_Y(y)$

Step 1: Use the CDF Method (Key Idea)

By definition:
$F_Y(y) = P(Y \leqslant y)$

Since $Y = g(X)$:
$F_Y(y) = P(g(X) \leqslant y)$

If g(.) is **monotonically increasing**, then:
$g(X) \leqslant y \Leftrightarrow X \leqslant g^{-1}(y)$

So:
$F_Y(y) = P(X \leqslant g^{-1}(y)) = F_x(g^{-1}(y))$

This is exactly what the slide shows using the graphs of $y = g(x)$ and $x = g^{-1}(y)$.

Step 2: Differentiate to Get the PDF

We know:
$f_Y(y) = d/dy\ [\ F_Y(y)\ ]$

So:

$f_Y(y) = d/dy\ [\ F_X(g^{-1}(y))\ ]$

Using the chain rule:

$f_Y(y) = f_X(g^{-1}(y)) \cdot d/dy\ [\ g^{-1}(y)\ ]$

Taking absolute value (to handle increasing or decreasing cases):

Final Formula:

$f_Y(y) = f_X(x) \cdot |\ dx/dy\ |$ evaluated at $x = g^{-1}(y)$

Slide 4: If g(x) is Decreasing

If g(.) is **monotonically decreasing**:

$F_Y(y) = P(Y \leqslant y) = P(g(X) \leqslant y)$

Now:

$g(X) \leqslant y \Leftrightarrow X \geqslant g^{-1}(y)$

So:

$F_Y(y) = P(X \geqslant g^{-1}(y)) = 1 - F_X(g^{-1}(y))$

Differentiate:

$f_Y(y) = -\ f_X(g^{-1}(y)) \cdot d/dy\ [\ g^{-1}(y)\ ]$

Taking absolute value again gives the **same general formula**:

$f_Y(y) = f_X(g^{-1}(y)) \cdot |\ dx/dy\ |$

So the **same formula works** for both increasing and decreasing functions.

Slide 5: Example 1 — Y = sin(πX / 2), X ~ Uniform(−1, 1)

Given

X is uniformly distributed on (−1, 1):

$f_X(x) = 1/2$, for $-1 < x < 1$
0 otherwise

Define:
$Y = \sin(\pi X / 2)$

We must find $\mathbf{f_Y(y)}$.

Step 1: Find the Inverse

$y = \sin(\pi x / 2)$

So:
$x = (2/\pi) \sin^{-1}(y)$

Step 2: Compute dx/dy

$dx/dy = (2/\pi) \cdot 1 / \sqrt{(1 - y^2)}$

Step 3: Apply the Transformation Formula

$f_Y(y) = f_x(x) \cdot |dx/dy|$

$= (1/2) \cdot (2/\pi) \cdot 1 / \sqrt{(1 - y^2)}$

$= 1 / (\pi \sqrt{(1 - y^2)})$

Domain:
Since sine output lies between −1 and 1:

Final Answer:
$f_Y(y) = 1 / (\pi \sqrt{(1 - y^2)})$, for $-1 < y < 1$
0 otherwise

The slide also sketches this **U-shaped curve**.

Slide 6: Case 2 — Function of Two Random Variables (Z = X + Y)

The slide asks to find:

(i) PDF of Z, $f_z(z)$

(ii) $f_z(z)$ if X and Y are independent

(iii) If $X \sim N(0,1)$, $Y \sim N(0,1)$, show $Z \sim N(0,2)$

(iv) If X, Y are exponential with parameter λ, find $f_z(z)$

Slide 7: CDF of Z = X + Y (Geometric View)

Start from definition:

$F_z(z) = P(Z \leqslant z) = P(X + Y \leqslant z)$

This means:

All (x, y) such that $x + y \leqslant z$

On the x–y plane, this is the **half-plane below the line**:

$y = z - x$

So:

$F_z(z) = \iint_{\{x + y \leqslant z\}} f_{X,Y}(x, y)\, dx\, dy$

The slide shows this shaded region clearly.

Slide 8: Leibniz Rule (Needed for Differentiation)

If:

$G(x) = \int$ from a(x) to b(x) of h(x, y) dy

Then:

$dG/dx = h(x, b(x)) \cdot db/dx - h(x, a(x)) \cdot da/dx + \int$ from a(x) to b(x) $\partial h/\partial x\, dy$

We use this because the **limits depend on z** in $F_z(z)$.

Slide 9: Deriving the PDF of Z

Start from:

$F_z(z) = \iint_{\{x + y \leqslant z\}} f_{X,Y}(x, y)\, dx\, dy$

Differentiate with respect to z using Leibniz rule:

Result:

$f_z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx$

This is the **general formula** for Z = X + Y.

Slide 10: If X and Y Are Independent (Convolution)

If X and Y are independent:

$f_{X,Y}(x, y) = f_x(x)\, f_y(y)$

So:

$f_z(z) = \int_{-\infty}^{\infty} f_x(x)\, f_y(z - x)\, dx$

This integral is called the **Convolution Integral**.

Slides 10–11: Special Case — Sum of Two Standard Gaussians

Given:

X ~ N(0,1), Y ~ N(0,1), independent

So:

$f_x(x) = (1/\sqrt{2\pi})\, e^{-x^2/2}$

$f_y(y) = (1/\sqrt{2\pi})\, e^{-y^2/2}$

Then:

$f_z(z) = \int (1/\sqrt{2\pi})\, e^{-x^2/2} \cdot (1/\sqrt{2\pi})\, e^{-(z-x)^2/2}\, dx$

By combining exponents and completing the square (as shown step-by-step in the slide), we get:

$f_z(z) = 1 / \sqrt{4\pi} \cdot e^{-z^2/4}$

This is a **Normal distribution** with:

Mean = 0

Variance = 2

So:

Z = X + Y ~ N(0, 2)

Slide 12: X and Y Are Exponential(λ)

Given:

$f_x(x) = \lambda e^{-\lambda x}, x \geqslant 0$

$f_y(y) = \lambda e^{-\lambda y}, y \geqslant 0$

Using convolution:

$f_z(z) = \int f_x(x) f_y(z - x) dx$

But constraints:

$x \geqslant 0$ and $z - x \geqslant 0 \Rightarrow 0 \leqslant x \leqslant z$

So:

$f_z(z) = \int_0^z \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda(z-x)} dx$

$= \lambda^2 e^{-\lambda z} \int_0^z dx$

$= \lambda^2 z e^{-\lambda z}, z \geqslant 0$

This is a **Gamma distribution (shape = 2)**.

Slide 13: Example — Z = X − Y

CDF:

$F_z(z) = P(Z \leqslant z) = P(X - Y \leqslant z)$

Rewrite:

$X - Y \leqslant z \Leftrightarrow Y \geqslant X - z$

So integrate the joint PDF over that region in the x–y plane (as shaded in the slide), then differentiate:

$f_z(z) = d/dz [F_z(z)]$

The **procedure is the same** as Z = X + Y, only the region changes.

Slide 14: Example — Z = X / Y and R = $\sqrt{(X^2 + Y^2)}$

Case 1: Z = X / Y

CDF:

$F_z(z) = P(X / Y \leqslant z)$

We must split into cases:

- If $Y > 0$: inequality stays the same
- If $Y < 0$: inequality reverses

So the plane is divided into two regions (as drawn in the slide), and integration is done **piecewise**.

Case 2: $R = \sqrt{(X^2 + Y^2)}$

Interpretation:

- (X, Y) is a random point in 2D
- R is the **distance from the origin**

So:
$F_R(r) = P(\sqrt{(X^2 + Y^2)} \leqslant r) = P(X^2 + Y^2 \leqslant r^2)$

This means:
Probability that (X, Y) lies **inside a circle of radius r**

So:
$F_R(r) = \iint$ over circle of radius r $f_{X,Y}(x, y)\, dx\, dy$

Then:
$f_R(r) = d/dr\, [\, F_R(r)\,]$

Final Summary

You learned:

- How to transform **one random variable** $Y = g(X)$ using the CDF method
- Main formula:
  $f_Y(y) = f_X(g^{-1}(y)) \cdot |dx/dy|$
- How to transform **two random variables** $Z = g(X, Y)$:
  - First find $F_z(z)$ by integrating over a region
  - Then differentiate using Leibniz rule
- For sum $Z = X + Y$:
  $f_z(z) = \int f_X(x)\, f_Y(z - x)\, dx$ (if independent → convolution)
- Applications covered:

- $Y = \sin(\pi X/2)$
- $Z = X + Y$ (general, Gaussian case, exponential case)
- $Z = X - Y$
- $Z = X / Y$
- $R = \sqrt{(X^2 + Y^2)}$

Formula Sheet (Quick Revision)

1. One-variable transformation:
   $f_Y(y) = f_X(g^{-1}(y)) \cdot |\, dx/dy \,|$
2. Two-variable transformation:
   $F_z(z) = \iint_{\{g(x,y) \leqslant z\}} f_{X,Y}(x,y)\, dx\, dy$
   $f_z(z) = d/dz\, [F_z(z)]$
3. Sum of RVs:
   $f_z(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z - x)\, dx$
   If independent:
   $f_z(z) = \int_{-\infty}^{\infty} f_X(x)\, f_Y(z - x)\, dx$ (Convolution)
4. Leibniz Rule:
   $d/dx \int_{a(x)}^{b(x)} h(x,y)\, dy$
   $= h(x,b(x))\, b'(x) - h(x,a(x))\, a'(x) + \int_{a(x)}^{b(x)} \partial h/\partial x\, dy$

If you want, I can now convert this into **clean LaTeX lecture-scribe notes**, or make **exam-focused short notes**, or solve **extra practice problems** from this topic step by step.